Apparent difficulty and measured model contribution move in opposite directions: a three-protocol calibration of negative-set construction across 94 ENCODE eCLIP datasets

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Abstract

Sequence models for RNA-binding proteins are evaluated against constructed negative windows, and how those windows are chosen is usually treated as a detail. Across 94 paired ENCODE eCLIP datasets we held the model, the positives, the chromosome-blocked folds and the estimator fixed, varied only how the negative windows were built, and measured each model's nested contribution over a 19-feature mono- and dinucleotide composition baseline refit under each protocol. Replacing GC-matched with dinucleotide-matched negatives lowered the model's own AUROC from 0.798 to 0.689 in 94 of 94 datasets while raising its measured contribution from 0.027 to 0.066. A bias-aware protocol drawing negatives from other proteins' binding sites gave the easiest discrimination of the three and the smallest contribution, 0.012. Apparent difficulty and measured contribution therefore move in opposite directions throughout this panel, spanning 5.4-fold for a 4-mer and between 3.8 and 7.6-fold across three model classes measured on identical rows; on the one independent benchmark we tested they move together, so the direction is a property of these three protocols and not a general law. Eight standard monotone reparameterisations reduced the 4-mer's span to a floor of 2.0 without closing it, and an exponent fitted to close it did not transport to an independent benchmark. Dividing by the baseline's headroom improved cross-protocol agreement on our own data but failed, out of sample, the falsification test we had specified for it in advance. We therefore recommend reporting the composition-only AUROC obtained under the same protocol alongside every headline AUROC, and not comparing contributions measured under different protocols.

Keywords: RNA-binding proteins; eCLIP; benchmarking; negative-set construction; sequence composition; incremental value; AUROC

1 Introduction

Sequence models that predict RNA-binding protein (RBP) occupancy are evaluated almost entirely by the area under the receiver operating characteristic curve (AUROC) on held-out windows.

Positive windows come from a crosslinking assay, most often eCLIP [2, 3], but negative windows have to be constructed, and the construction is usually given a sentence in the methods. Where it is specified, negatives are commonly unbound regions of the same transcripts [20] or a composition-matched genomic sample [6]. Horlacher *et al.* [4] benchmarked eleven RBP prediction methods across hundreds of CLIP-seq experiments under two negative-set definitions and showed that a method's apparent performance depends on which definition is used. They proposed a bias-aware alternative in which negatives are drawn from the binding sites of other RBPs rather than from genomic background. The same dependence has since been reported for transcription factor binding sites, where six sampling strategies were compared against a position weight matrix baseline and dinucleotide-shuffled negatives performed worst of those tested [5].

How large that dependence is, and in which direction it acts, remains open. A raw AUROC conflates the signal a model found with the difficulty of the discrimination it was set, and whoever builds the benchmark controls the second. Negatives sampled uniformly from the genome differ from crosslinked positives in base composition, so a classifier responding to nucleotide frequency alone will score well without having learned anything sequence-specific. Matching negatives on GC content is the usual mitigation, but GC content is one linear functional of the sixteen-cell dinucleotide simplex, leaving the remaining fourteen degrees of freedom of compositional difference available to any model. Comparable circularity is characterised in variant-effect prediction, where the assembly of the evaluation set both inflates apparent accuracy and reorders methods [10], and the sensitivity of genomic machine-learning evaluations to the definition of the negative class is recognised more generally [7, 8, 11]. That a difference in AUROC between two models depends on how strong the baseline model is has also been characterised outside genomics [15].

We call a model's own AUROC on a given protocol its *apparent* AUROC, since it is the number a benchmark reports and it depends on the protocol as much as on the model. A model's increment over an explicit composition baseline is less exposed to compositional confounding than its apparent AUROC. We define the nested contribution as the difference between the pooled out-of-fold AUROC of a logistic model on composition features together with the model's score, and that of the composition features alone, fitted on the same rows and the same folds. Every contribution reported here is therefore an increment over an order-two baseline, the mono- and dinucleotide frequencies plus entropy. That choice is not innocuous, and the Discussion measures what a third order costs.

We assembled 94 ENCODE eCLIP datasets and constructed negatives three ways: matched on GC content, matched on all sixteen dinucleotide frequencies, and drawn from the binding sites of other RBPs assayed in the same cell line, following Horlacher et al. [4]. The three protocols differ in what they hold fixed, and not only in how tightly they match composition: the two composition-matched protocols also draw candidates of the same transcript region class, which the bias-aware protocol does not, and only the bias-aware protocol's negatives are transcribed RNA. Model, positives, chromosome-blocked folds and estimator were held fixed across arms, and the composition baseline was refit on each arm's own windows so that no baseline was carried between designs. Three model classes were measured under all three protocols: a 4-mer logistic regression, a

7089-parameter convolutional network, and a fully fine-tuned 19.78 M-parameter nucleotide language model.

Apparent difficulty and measured contribution move in opposite directions. Dinucleotide matching reduced apparent AUROC in all 94 datasets while raising the nested contribution by a factor of 2.50 in panel means, and the bias-aware protocol gave both the easiest discrimination of the three and the smallest contribution. The contribution spanned 5.4-fold across protocols for the 4-mer, and between 3.8 and 7.6-fold across the three model classes. For scale, on the dinucleotide arm of this panel 0.019 AUROC separates the 4-mer from the convolutional network and 0.122 separates it from SpliceBERT, so the protocol moves the measured quantity by nearly three times the smaller of those two steps and by about half the larger. Eight standard monotone reparameterisations reduced the span without closing it, and an exponent fitted to close it did not transport to an independent benchmark. On that basis we recommend reporting the composition-only AUROC alongside every headline AUROC, and we report where that helps and where it does not.

2 Materials and Methods

2.1 Panel selection

Candidate experiments were every released ENCODE eCLIP experiment in K562 and HepG2 (139 and 105 respectively). Where a protein had several peak files we took the one with the most biological replicates, breaking ties by ENCODE's `preferred_default` flag, and required `output_type=peaks`, `assembly=GRCh38`, `status=released` and `file_format=bed`. Every candidate was preprocessed under both composition-matching protocols; datasets retaining at least 400 out-of-fold scored pairs were eligible.

The study panel was then defined once by the dinucleotide arm, which is the stricter control: eligible datasets were sorted by pair count with a stable sort and every second one retained, giving 95 datasets over 79 proteins. We sampled systematically by rank instead of taking the largest N, because AUROC correlates with dataset size in these data, negligibly for the composition baseline (Pearson r on $\log_{10} n$ of -0.04 in the GC arm) but strongly for the models, reaching 0.70 for SpliceBERT under dinucleotide matching, so a size-truncated panel would confound panel membership with the measured quantity; the retained panel spans the 0th to 99th percentile of the candidate size distribution. Of the 95, 94 also cleared the floor in the GC arm and constitute the paired panel used for every three-protocol result, giving 282 protocol-dataset cells. All 95 are used only for the comparison of standalone model AUROCs quoted in the Introduction, which needs no GC-arm counterpart. Panel membership was written once to a manifest that every downstream stage reads. Supplementary Table S1 lists every dataset with its ENCFF file and ENCSR experiment accession.

Sixteen proteins are assayed in both cell lines and contribute two datasets each, fifteen of them in the paired panel. All headline intervals therefore resample proteins rather than datasets.

2.2 Windows

Positive windows are 101 nt centred on the midpoint of each peak interval. Peak width and the significance columns are unused: we applied no further p-value, q-value or fold-change filter, because ENCODE's released `peaks` files are already thresholded and across our 95 files the minimum log2 fold-enrichment is 3.0000 and the minimum $-\log_{10} p$ is 3.0003, which are the criteria of Van Nostrand et al. [3]; 91.7% of the 463,091 peaks carry ENCODE's IDR label. The positive set is therefore every peak ENCODE called and no weaker ones. Minus-strand windows are reverse-complemented and transcribed to RNA, so every sequence is the strand the protein sees. Windows containing an ambiguous base, falling outside the assembly, duplicating an existing window by start coordinate, or carrying no region annotation are dropped and counted. Analysis is restricted to chr1 to chr22 and chrX; chrY and all scaffolds and alternates are excluded.

Region class is assigned from GENCODE v45 [9] as the first of (5′UTR, 3′UTR, CDS, non-coding exon, intron) whose interval set contains the window midpoint. Region intervals were built from all transcripts with no gene-type filter and merged, with strand disregarded at this step; introns were computed as the gaps between consecutive exons. GTF 1-based inclusive coordinates are converted to 0-based half-open.

2.3 The three negative-set protocols

Each protocol produces one negative per positive. All three use a single random stream seeded at 7 per protein.

GC-matched. Candidates are windows of the same region class on the same chromosome, at least 500 nt from any peak of the target protein, drawn with probability proportional to the length of each free interval. A candidate is accepted if its GC content is within 0.05 of the positive's. The matcher relaxes: after 25 unsuccessful draws the tolerance doubles to 0.10, and if 40 draws fail the best candidate seen is accepted provided it is within three times the nominal tolerance, that is 0.15. Only the target protein's own peaks are excluded, so a GC-matched or dinucleotide-matched negative may be another protein's binding site. The negative inherits the positive's strand label; the consequences are quantified in the Results.

Dinucleotide-matched. Positives are grouped by region and chromosome, and for each group a candidate pool of max(1500, $8n$) windows is sampled. For each positive the nearest unused candidate is assigned greedily under the L_1 distance over the sixteen dinucleotide *counts*, searched with a k-d tree over the forty nearest neighbours. The distance is over counts, not frequencies, because integer distances are exactly representable, and each candidate carries an additional tiebreak column that increases strictly with genomic position and is bounded below 0.5, so it can never reorder candidates whose integer distances differ but resolves exact ties by position. Without both devices the choice among tied candidates depends on the floating-point behaviour of the k-d tree implementation and a different negative is selected for about 8% of pairs between CPU architectures. Assignment is greedy, not optimal; achieved distances are reported so the quality of the approximation is visible. Used coordinates are tracked across all groups, because one interval can be annotated as both a non-coding exon and a 3′UTR and so appear in two region pools. No maximum distance is imposed.

Bias-aware (other RBPs' sites). Following Horlacher et al. [4], negatives for a target are the positive windows of the other panel proteins assayed in the same cell line, sampled 1:1 without replacement, excluding any donor window within 500 nt of any of the target's own positive windows. Our version deviates from Horlacher et al. [4] in two ways: our donor pool is the study panel (40 to 48 donor proteins per dataset, median 45) and not a full ENCODE survey, and exclusion is by distance from the target's 101 nt windows, not from its full peak intervals. Both the donor pool and the target's positives are read from the GC arm's window tables, so this arm inherits the GC matcher's retained positive set and its pair count is identical to that arm's. Sampling is performed within fold, so every pair stays inside one cross-validation fold. These negatives are transcribed, accessible to crosslinking and strand-correct by construction, and are not composition-matched in any respect.

What each protocol holds fixed. The three protocols are not three settings of one knob, and the difference changes how their results should be read. Both composition-matched protocols draw candidates of the same transcript region class as the positive, so the region label carries no information about the class. The bias-aware protocol matches fold only, which leaves region free; how far region separates the two classes there is measured in Section 3.2. Only the bias-aware protocol's negatives are transcribed RNA, and conversely 67.6% of GC-arm negatives lie in loci not transcribed on the strand the window is labelled with. Section 3.2 bounds what the region asymmetry contributes; the strand and transcription placebos in the Results bound the other two.

2.4 Achieved match quality

Achieved matching was quantified over every pair in each arm, 456,734 in the GC and bias-aware arms and 457,998 in the dinucleotide arm (Table 1).

Table 1: Achieved match quality between each positive and its assigned negative, all pairs. Columns two to five describe the absolute GC difference between the pair; L_1 is over the sixteen dinucleotide frequencies, on a 0 to 2 scale.

arm	med. $\lvert\Delta\mathrm{GC}\rvert$	p90	p99	max	within 0.05	med. L_1
GC-matched	0.0297	0.0495	0.1089	0.1486	94.8%	0.500
dinucleotide-matched	0.0198	0.0693	0.1485	0.4159	85.0%	0.220
bias-aware	0.1387	0.3169	0.4555	0.6733	20.4%	0.700

The GC matcher holds its nominal tolerance for 94.8% of pairs and its observed maximum, 0.1486, sits at the 0.15 fallback cap. "Dinucleotide-matched" likewise means a median L_1 of 0.220 rather than zero: the matcher improves composition distance 2.27-fold relative to the GC arm but does not eliminate it. The bias-aware arm is not composition-matched at all, with a median GC gap 4.7 times the GC arm's.

2.5 Cross-validation

Evaluation is grouped 5-fold cross-validation over whole chromosomes. The chromosome-to-fold map is frozen once for all datasets in both cell lines, so the fifteen proteins measured in both lines are evaluated on the same chromosomes and a between-line difference cannot be a partition artefact. The map was chosen by random-restart hill climbing on peak counts only, never on labels or model output, minimising the summed squared deviation from equal mass per fold with every dataset weighted equally; it was required to beat round-robin assignment, to hold the median per-dataset maximum deviation at or below 0.05, and to place no more than half of any dataset's data in one fold. Fold i is the test fold, fold $i + 1 \bmod 5$ is validation and the remaining three train, so each fold is test once, validation once and training three times, and no additional data is spent on model selection.

Every window is scored exactly once out of fold and all AUROCs are computed on the pooled out-of-fold score vector. Pooling, instead of averaging per-fold AUROCs, makes the estimate invariant to fold size; the least balanced dataset places 0.42 of its data in a single fold.

Leakage between folds was audited by exact 32-mer sharing, not by clustered sequence identity. A 101 nt window contains 70 distinct 32-mers, and two unrelated sequences share one with negligible probability, so a shared 32-mer indicates common origin. Over all 94 datasets, holding fold 0 out, a median of 2.0% of held-out windows shared any 32-mer with a training window (maximum 24.3%), a median of 0.08% shared more than half of their 32-mers, and the median dataset contained no window sharing more than 90%. Chromosome grouping therefore leaves a small residue of homologous sequence across folds, which is the quantity a sequence-identity clustering step would remove.

2.6 The composition baseline and the nested contribution

The composition baseline is a 19-column design matrix: three mononucleotide frequencies, fifteen dinucleotide frequencies and the Shannon entropy of the window's base composition. One level is dropped from each frequency family because frequencies sum to one and retaining all of them makes the design singular; dropping a level does not change the space the block spans. Every column is standardised over the whole dataset, not within fold; the transform is label-free and identical in both terms of every comparison, so it cannot manufacture a contribution. GC content is not a feature: it is the spanned combination C+G, and mononucleotide counts are almost, but not exactly, marginals of dinucleotide counts: over a window of length L the dinucleotide counts sum to $L - 1$ and recover each base's count only up to whether the terminal base is that base. The linear block therefore has rank 18 and not 15, with a median 0.14% of each mononucleotide column's variance lying outside the span of the dinucleotide columns (maximum 0.25% over 25 datasets), and the entropy column adds a nineteenth, nonlinear direction. What the two protocols constrain is nonetheless asymmetric by construction, and that asymmetry is what the design argument rests on: GC matching fixes a single linear functional of local composition, whereas dinucleotide matching fixes all fifteen degrees of freedom of the dinucleotide simplex, so it leaves the baseline strictly less compositional variation to exploit. The baseline is refit on each protocol's own windows and is never carried between protocols.

The nested contribution of a model is

$$\text{AUROC}(\text{composition} + \text{standardised model score}) - \text{AUROC}(\text{composition alone}).$$

Both terms were fitted by L_2-penalised logistic regression ($C = 1$, lbfgs, at most 3000 iterations), once per fold on that fold's training rows, and evaluated as pooled out-of-fold linear predictors on identical rows and folds. The model-score column is the out-of-fold score for every row, training rows included, so the nested model is never fitted on an in-sample score; had it been, the fitted and evaluated covariate would sit on different scales and the mismatch would grow with how much the underlying model overfits, which differs sharply between a 256-column penalised logistic and a fully fine-tuned transformer. The two were compared by DeLong's paired estimator [12], in the $O(n \log n)$ midrank form of Sun and Xu [13]. The paired estimator is necessary here because the two score vectors are fitted on overlapping training data and evaluated on identical rows, so they are strongly correlated and treating them as independent would overstate the variance of their difference: the median DeLong standard error of the difference is 0.26 times what independence gives. Confidence intervals on a single dataset's contribution are Wald intervals on the DeLong standard error.

Demler et al. [14] show that DeLong's test is not valid for strictly nested models, because under the null the added predictor's contribution degenerates. Two things bound the exposure here. The panel-level intervals, which carry every headline, do not use the DeLong standard error at all; they are protein-clustered bootstrap intervals over per-dataset point estimates. And where the DeLong standard error is used, for the per-dataset significance counts, it is conservative relative to the recommended alternative: a penalised likelihood-ratio test on the same fits calls 89 of 94 datasets significant in the GC arm against DeLong's 80. A third feature of the design might have cut the other way and does not: negatives are matched 1:1 to positives, so the rows are paired, while DeLong treats the two classes as independent samples. Resampling matched pairs instead reproduces the DeLong standard error to within 1% (median ratio 0.99 over 30 datasets), so the matching needs no correction.

Both arms are evaluated out of fold. An in-sample composition AUROC compared against an out-of-fold model AUROC flatters composition sufficiently to reverse the comparison on individual datasets.

2.7 Models

4-mer logistic regression. Raw 4-mer counts over the RNA sequence (98 per window, 256 columns) into L_2-penalised logistic regression ($C = 1$). One model per fold, out-of-fold score taken as the linear predictor. The k-mer order is 4, and the contrast is positive at every order from 3 to 6.

Convolutional network. A DeepBind-style architecture [19]: convolution (4 to 16 channels, width 12), ReLU, max-pool 4, convolution (16 to 32, width 8), ReLU, global max-pool over position, dense 32 to 64, ReLU, dropout 0.5, dense 64 to 1. 7089 parameters, all trained from scratch. Global max-pooling makes the model position-invariant, which is necessary here because exploratory analysis placed the discriminative signal approximately 15 nt off centre, varying by protein.

SpliceBERT. The pretrained nucleotide language model of Chen et al. [21], `multimolecule/splicebert`, fully fine-tuned; 19.78 M parameters, all trainable. The classification head is a mean pool over real nucleotide tokens, masking classification, separator and padding tokens, then dense to 128, ReLU, dropout 0.3, dense to 1. Weights are baked into the container image at build time.

Both neural models: AdamW, weight decay 0.01, batch size 32, binary cross-entropy, single precision, at most 12 epochs with early stopping on validation AUROC at patience 4, best checkpoint restored before test scoring. Learning rate 10^{-3} for the head and 3×10^{-5} for a fine-tuned encoder. Nominal epochs overstate the training performed: the median best epoch for SpliceBERT was 3 and most runs stopped by epoch 6 or 7, whereas the convolutional network typically ran the full 12. Initial weights were drawn before the seed was set, so initialisation is unseeded across all 2830 committed fold-runs, that is two models by five folds by 94 datasets in the GC and bias-aware arms and 95 in the dinucleotide arm; the measured cost is a per-dataset standard deviation of 0.006 to 0.010 in the nested contribution, inducing about 0.001 on a 94-dataset mean against a reported interval half-width of about 0.008. The convolutional and transformer arms were trained on different accelerators, which changes floating-point summation order and is covered by the same measurement.

2.8 Inference

Because fifteen of the 79 proteins contribute two datasets each and the within-protein correlation of the primary contrast is 0.92, every headline interval is a percentile interval from a bootstrap that resamples *proteins* and takes all datasets of each sampled protein, 4000 draws. Dataset-level intervals are 1.05 to 1.23 times narrower and are not reported as headline values.

Per-dataset significance counts are also reported at a design effect of 1.35 applied to the DeLong standard error, which is intended to cover two things the estimator does not: that it conditions on two score vectors as fixed when both are fitted, and that it treats spatially clustered windows as independent. We measured both components instead of assuming them (`scripts/design_effect.py`). Resampling whole genomic blocks inflates the standard error of the contribution by a median factor of 1.100, and the factor is insensitive to block length, moving only to 1.104 at 100 kb and 1.138 at 1 Mb, so the objection that a 10 kb block is smaller than a gene does not bite. Permuting the model score within fold and refitting gives 1.048. The measured product is therefore 1.15 rather than 1.35. We retain 1.35, because it is the more conservative of the two and because the count it produces was fixed before the components were measured; a reader who prefers the measured figure should read 75 of 94 rather than 72 of 94 below. This is one of two quantities in the paper that the verification harness cannot check from the released evidence, because the block bootstrap needs the genomic coordinates that the published per-window score files do not carry.

2.9 Reproducibility

All published values are reproduced offline by a single command against committed evidence. The harness runs 696 numeric assertions against a frozen expectations file, and the harness asserts that no fewer than that number ran, so a check cannot silently skip. Two assertions are stronger than regression gates: 285 published AUROCs are rebuilt from committed per-example scores to a maximum absolute difference of 2.2×10^{-16}, and the headline contrast is rebuilt from raw sequence to 1.2×10^{-6}. Per-window out-of-fold scores for all three model classes are committed, so every cell of the model-class comparison is recomputable from the repository alone, and is not asserted against a hand-entered table.

Three devices guard against platform-dependent results: dinucleotide distances are computed on integer counts, candidate ties are broken by genomic position, and the panel sort is stable. Each was added after an observed discrepancy between CPU architectures or between runs.

Analyses used NumPy [22], SciPy [23], scikit-learn [24], PyTorch [25] and Matplotlib [26].

3 Results

3.1 The negative-set protocol changes measured contribution more than five-fold

Across 94 paired datasets the same 4-mer model gave three different accounts of its own contribution depending only on how the negative windows were built (Table 2, Figure 1).

Table 2: Composition-only AUROC, the 4-mer's own AUROC, the AUROC of the two together, and the nested contribution, under three negative-set protocols. The nested contribution is the composition-plus-4-mer column minus the composition-alone column by construction. The 4-mer's own AUROC is what a study would report as its headline. Model, folds and estimator are identical across arms; positives are identical in 10 of 94 datasets and share a median Jaccard of 0.997, as the Limitations set out. $n = 94$ datasets.

protocol	composition	4-mer alone	composition + 4-mer	contribution
dinucleotide-matched	0.6274	0.6879	0.6937	+0.0663
GC-matched	0.7827	0.7981	0.8092	+0.0265
bias-aware	0.8248	0.8169	0.8370	+0.0122

Replacing GC-matched with dinucleotide-matched negatives reduced the 4-mer's own AUROC in all 94 datasets (paired Wilcoxon $p < 10^{-15}$), from 0.7981 to 0.6879, while raising the nested contribution from +0.0265 to +0.0663. The contrast in contribution was +0.0398 (95% CI +0.0325 to +0.0477, protein-clustered; positive in 88 of 94 datasets). Evaluated by its own AUROC the model appears to have deteriorated; evaluated by contribution it appears to have improved by a factor of 2.50 in panel means, or 2.94 as the geometric mean of the 88 per-dataset ratios that are defined. Across the three arms the contribution spanned 5.42-fold (95% CI 4.43 to 6.58, protein-clustered).

The bias-aware protocol did not behave as we predicted. Because both classes are genuine crosslink sites, we expected composition alone to fall towards 0.5 and the contribution to rise. Neither occurred: composition alone was the highest of the three arms at 0.8248, the AUROC of composition plus the model was the highest at 0.8370, and the contribution was the lowest at +0.0122, or 0.53 times the GC arm (geometric mean over the 83 datasets positive in both arms; the ratio of panel means is 0.46). Part of this is transcript-region mix and not site composition; the next subsection separates the two.

The two AUROC rankings of the three protocols agree with each other, and the ranking by contribution is their exact reverse. Harder discriminations do reveal more of what a model adds; what does not follow is that a bias-aware protocol is the harder one. The starkest form of this is that the 19-feature composition baseline beat the 4-mer outright in 14 of 94 datasets under dinucleotide matching, 29 of 94 under GC matching, and 53 of 94 under the bias-aware protocol, where it also beat the model on the panel mean. Under GC matching the contribution was significantly positive in 80 of 94 datasets on unadjusted marginal intervals, 79 of 94 after Benjamini-Hochberg control within the panel, and 72 of 94 once the design effect is applied (Methods).

The direction of the GC-versus-dinucleotide contrast is implied by the design. The composition baseline reads local composition, and the two protocols constrain it asymmetrically: GC matching fixes a single linear functional of that composition, whereas dinucleotide matching fixes all fifteen degrees of freedom of the dinucleotide simplex, so the arm that constrains more must leave the baseline weaker. Only the magnitude of that contrast is informative. The position of the bias-aware arm is not implied by the same argument. That protocol constrains none of those degrees of freedom, and its negatives are not composition-matched in any respect (median absolute GC difference 0.1387, against 0.0297 for the GC arm). It nonetheless has the highest composition baseline of the three.

We bounded two artefacts against placebos matched on region and GC, with twenty placebo seeds per dataset per arm. Both tests are on the GC-versus-dinucleotide contrast and on subpanels, and each percentage below is taken against its own subpanel's contrast and not against the +0.0398 headline. Strand misassignment, which arises because a matched negative inherits its positive's strand label, accounted for −0.0055 (CI −0.0089 to −0.0022) of the +0.0378 contrast on the 40 datasets where the test is possible, leaving 85.4%. Negatives drawn from untranscribed loci accounted for −0.0043 (CI −0.0069 to −0.0018) of the +0.0414 contrast on the 84 datasets with expression data, leaving 89.7%. The two mechanisms are correlated, so 0.854 and 0.897 should not be multiplied. On the transcription control, 40.1% of negatives lie in untranscribed loci and 67.6% are not transcribed on the strand the window is labelled with, but both fractions are balanced between the two arms, the first to within 1.2×10^{-5} ($p = 0.64$), and a confound present equally in both arms cannot produce a difference between them. Counting all negatives, 42.8% of GC-matched negatives carry the strand of their own gene against 47.4% in the dinucleotide arm, which is the higher of the two in 37 of 40 datasets ($p = 5.0 \times 10^{-9}$), so the spurious directional cue is stronger in the arm with the smaller contribution and acts against the reported direction.

The strand test admits a stronger reading, which we report here so that it is not left to be found. Within the panel, regressing the contrast on the sense-strand fraction gives a slope of -0.187, and extrapolating to a fully sense-correct arm gives -0.046, a sign reversal. That extrapolation carries no weight: over the negatives whose strand is determinable, the fraction spans only 0.433 to 0.615 across datasets, so the leverage available is 0.18 and the extrapolation runs far outside it, the interval on the extrapolated value is -0.189 to $+0.072$, and the direct test, the correlation between the sense fraction and the contrast, is -0.240 ($p = 0.136$) and null. The point estimate is nonetheless the one number in this paper whose naive reading reverses the headline. None of these three controls covers the three-arm ordering, only the composition-matched pair; Section 3.2 is the corresponding bound for the bias-aware arm.

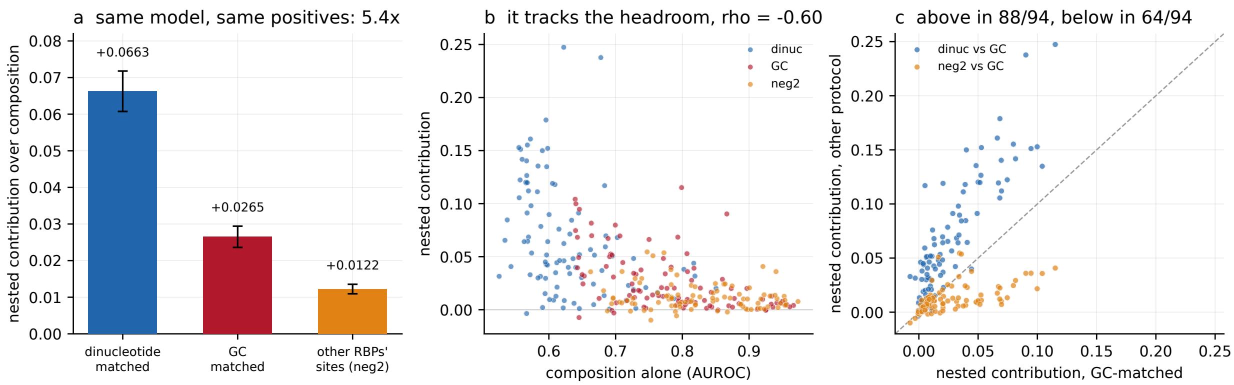

Figure 1: Nested contribution of a 4-mer model under three negative-set protocols, $n = 94$ datasets. (a) Panel means by protocol, $+0.0663$, $+0.0265$ and $+0.0122$, a 5.4-fold span for one model on one set of positives; bars are standard errors over datasets. (b) The same contributions against the composition-only AUROC each protocol leaves, one point per protocol-dataset cell, coloured by protocol (pooled Spearman -0.60): the contribution tracks the headroom. (c) Per dataset, the contribution under each of the other two protocols against its value under GC matching. Dinucleotide matching lies above the diagonal in 88 of 94 datasets and the bias-aware protocol below it in 64 of 94.

3.2 Part of the bias-aware arm's baseline is transcript region, and the ordering survives removing it

The bias-aware arm's position is the one result here that the design does not imply, so it is the one most exposed to a design asymmetry, and there is one. Both composition-matched protocols draw candidates of the same region class as the positive, so region carries nothing: a classifier given only the five region dummies scores exactly 0.5000 in both arms, in all 94 datasets. The bias-aware protocol matches fold only, and there region alone separates the two classes at a median AUROC of 0.748, above 0.70 in 67 of 94 datasets, with the positive and negative region marginals differing at a median total variation distance of 0.419 (Table 3). The effect is graded, not uniform: the more region separates in a dataset, the more its bias-aware baseline exceeds its GC baseline (Spearman +0.287, $p = 0.005$) and the larger its contribution deficit against that arm (Spearman -0.253, $p = 0.014$).

We therefore rebuilt the arm with region matched, stratifying the donor draw on region class instead of drawing uniformly within the fold. Every other element of the construction is unchanged, and the rebuild keeps all 456,734 pairs, so no pair is lost to subsampling. It achieves the match exactly, at a region-only AUROC of 0.5000 and a total variation distance of 0.0000. The composition baseline falls from 0.8248 to 0.8052 and the contribution from +0.0122 to +0.0092, so 46% of the arm's baseline excess over the GC arm is region mix and the rest is not. The ordering is unchanged: the region-matched arm still carries the highest composition baseline of the three and still yields the lowest contribution, and the span against the dinucleotide arm widens from 5.42 to 7.20.

Both alternative constructions agree. Reweighting the donors already drawn, instead of redrawing them, gives 0.8017 and +0.0062 from a construction that discards 30% of the rows. And on the half of the panel where region separates least, which requires no rebuild at all, the bias-aware baseline is still the higher (0.8184 against 0.7962 for the same datasets' GC arm) and the contribution ratio is still below one at 0.641.

We do not propose this arm as a fourth protocol. Horlacher et al. [4]'s negative-2 does not match region either, so a region-matched version of it is not the design the field proposed. Its purpose is to measure how much of the arm's baseline is transcript annotation and how much is binding-site composition.

Table 3: What the region label alone achieves in each arm, and the bias-aware arm rebuilt with region matched. Region is matched by construction in both composition-matched arms and free in the bias-aware arm. The rebuilt arm is a diagnostic, not a fourth protocol; the reweighted row is a second estimate from a different construction, which discards 30% of the rows. $n = 94$ datasets.

arm	pairs kept	region AUROC	region TV	composition	contribution
dinucleotide-matched	100%	0.5000	0.0000	0.6274	+0.0663
GC-matched	100%	0.5000	0.0000	0.7827	+0.0265
bias-aware	100%	0.7484	0.4192	0.8248	+0.0122
bias-aware, region-matched	100%	0.5000	0.0000	0.8052	+0.0092
the same, by reweighting	70%	0.5026	0.0043	0.8017	+0.0062

The explanation for the arm's position is therefore narrower than that distinct RBPs occupy compositionally distinct sites. Distinct RBPs occupy distinct transcript regions, which the composition baseline reads through the base composition characteristic of each region, and they also occupy compositionally distinct sites within a region. Both contribute, and only the second is about binding-site composition.

3.3 No standard reparameterisation removes the protocol dependence

AUROC is compressive near unity, so a contribution measured against a baseline of 0.82 is arithmetically smaller than the same contribution measured against 0.63, and part of the 5.42-fold span comes from the scale itself (Figure 2).

We compared eight rescalings of the same 282 protocol-dataset cells. Seven are monotone transformations of the contribution itself; the eighth, division by the baseline's headroom, is not a transformation of the contribution at all but a normalisation by a second quantity, and it is kept in the same comparison because it is what a reader would reach for. The span was reduced but not eliminated by any of them. Somers' D [16], an affine map of AUROC, returns 5.42 by construction and serves as an internal control. Links that stretch the upper tail reduced the span in proportion to how far they stretch it: arcsine 3.99, complementary log-log 3.94, binormal d' 3.18, logit 2.72. The smallest span, 2.00 (CI 1.67 to 2.46), was obtained by dividing the contribution by the headroom of its own baseline, 1 minus the composition AUROC. Normalising by the excess over chance instead increased the span to 18.2, because that denominator varies with protocol and approaches zero on some datasets.

A span of 2.00 should not be compared against unity. The ratio of the largest to the smallest of three sample means is bounded below by one and is upward-biased, so some span is expected under any sampling. Simulating equal true arm means while preserving the observed between-arm dataset pairing gave a null distribution with median 1.064 and 95th percentile 1.175, so the residual span lies outside what sampling alone produces.

Division by headroom belongs to a family of baseline-dependent normalisations, the $p = 1$ member of $g/(1-c)^p$, where g is the contribution and c the composition AUROC, and the minimum of the sweep above falls on the $p = 1$ member. Over the family the span falls to 1.004 at $p = 1.544$ (Table 4), so a monotone rescaling that equalises these three protocols does exist.

Table 4: Fold span across the three protocols for $g/(1-c)^p$, where g is the nested contribution and c the composition-only AUROC. An exponent that equalises the three protocols exists in sample.

p	0	0.5	1.0	1.25	1.544	1.75	2.0
span	5.42	3.43	2.00	1.48	1.004	1.34	1.95

That exponent describes this benchmark only. Fitted to the two negative sets of Horlacher et al. [4] the equalising exponent is 3.649, and our exponent leaves their benchmark at 2.340 against the 2.381 it started from. A normalisation strong enough to equalise one benchmark therefore has to be refitted on the next.

Extending the sweep to the other two model classes qualifies the headline figure. The floor over the eight reparameterisations is 2.00 for the 4-mer, 2.91 for the convolutional network and 1.68 for SpliceBERT: no model class reaches protocol independence among fixed coordinates and all three exceed the equal-means null, but 2.00 is the 4-mer's floor, and each model class has its own. The fitted exponent also performs worse for the neural models, reaching 1.213 for the convolutional network and 1.124 for SpliceBERT against 1.004 for the 4-mer, and the equalising exponent itself differs by a factor of 1.22 between model classes measured on the same benchmark (1.544, 1.843, 1.507).

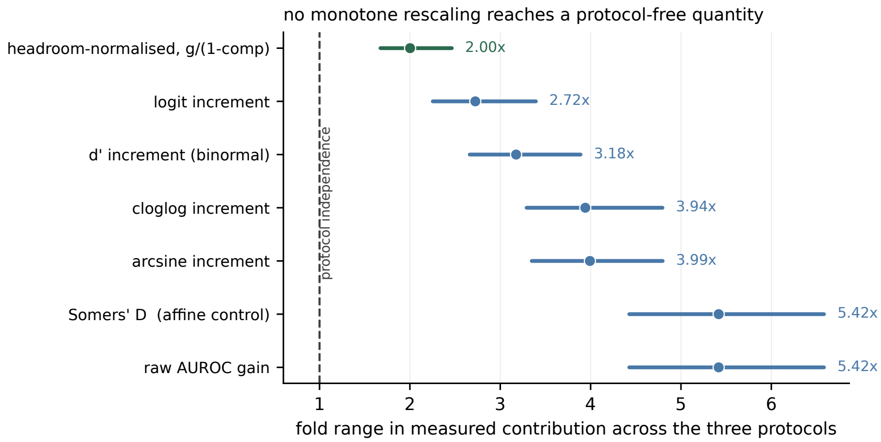

Figure 2: Fold span in nested contribution across the three protocols under eight monotone rescalings of the same 282 protocol-dataset cells. Seven of the eight are shown; normalising by the excess over chance is omitted because its span of 18.2 is off scale. Points are the span and bars are 95% percentile intervals from a bootstrap over the 79 proteins (4000 draws); the dashed line marks protocol independence. Somers' D is an affine map of AUROC and returns the raw value by construction, serving as an internal control. The smallest span, 2.00 (1.67–2.46), is obtained by dividing by the headroom of the baseline.

3.4 The composition baseline explains more of the dependence than the protocol label, for a k-mer model

Pooled over all 282 protocol-dataset cells the contribution was inversely related to the composition baseline (Spearman -0.600, $p = 6 \times 10^{-29}$), so much of what a protocol does is set how much room the baseline leaves (Figure 3).

Regressing contribution on a quadratic in the composition baseline over all 282 cells and then adding protocol indicators, the protocol label contributed 1.0% of variance given the baseline and the baseline contributed 11.0% given the label. Neither increment is corrupted by collinearity, since the baseline recovers the arm with only 58.5% accuracy against a chance rate of 33.3% and the protocol dummies regressed on the baseline curve give variance inflation factors of 1.11 and 1.32. The two increments answer different questions, though, and their ratio is not a decomposition of one variance into two competing causes: the first is a protocol- specific offset after a smooth baseline curve, the second a pooled within-arm gradient. The protocol's actual manipulation is within dataset, where the dinucleotide baseline is below the GC baseline in 94 of 94 datasets, so the first increment is estimated almost entirely from between-dataset variation. With dataset fixed effects the two increments become 0.33% and 6.33%.

The bias-aware protocol usually raises the baseline relative to GC matching, but lowers it in

27 of 94 datasets, which permits a within-panel test. Where it raised the baseline its contribution deficit was -0.0212 (CI -0.0269 to -0.0157); where it lowered the baseline the deficit was abolished and the point estimate changed sign, at $+0.0028$ (CI 0.0000 to $+0.0063$, a dataset bootstrap; the protein-clustered interval is $+0.0002$ to $+0.0063$). Within datasets, the change in baseline and the change in contribution were related at Spearman -0.664 ($p = 3 \times 10^{-13}$).

Matching on the baseline directly is possible only across datasets, because the dinucleotide baseline is below the GC baseline in all 94 datasets and the two are therefore perfectly rank-confounded within a dataset. Under nearest-baseline matching, with a caliper of 0.02 AUROC, the $+0.0398$ contrast fell to -0.0087 (CI -0.0265 to $+0.0122$, $n = 41$ pairs; the interval is -0.0284 to $+0.0219$ when the bootstrap is clustered on the reused GC partners). This is not a clean test of the contrast, for two reasons that point the same way. The retained pairs are drawn from opposing tails of their respective distributions: mean baseline 0.6875 against a panel mean of 0.6274 for the dinucleotide members, and 0.6904 against 0.7827 for their GC partners. The 41 comparisons are served by 24 distinct GC partners, one of them reused eleven times, and proteins are not matched; the common support for this pair is 0.073 AUROC wide over 33 of 188 cells. More fundamentally, the composition baseline is the mediator this paper proposes for the protocol's effect, so matching on it estimates a controlled direct effect, and a null direct effect is what the mechanism predicts, not evidence against it.

The GC-versus-bias-aware pair overlaps by 0.212 AUROC over 130 of 188 cells, where the same design is better conditioned. There, with a caliper of 0.01, the bias-aware arm retained a deficit of -0.0081 (CI -0.0130 to -0.0036) under nearest-baseline matching and -0.0043 (CI -0.0077 to -0.0012) as a within-dataset intercept at zero baseline shift. Decomposed by pair, the protocol label contributed 0.05% of incremental variance for GC versus dinucleotide and 3.40% for GC versus bias-aware, so the pooled 1.0% averages a comparison that is not identified with one that is.

Within arms, the relation between baseline and contribution was present for composition-matched negatives and absent otherwise: Spearman -0.545 (GC), -0.462 (dinucleotide) and -0.122 (bias-aware, not significant). This suggests two protocol families and not three interchangeable protocols, with a residual of 0.004 to 0.008 AUROC between families. The statistic correlates a difference with its own subtrahend, which is legitimate but not invariant to which term is placed on the horizontal axis, and one arm does not survive the change. Putting the composition-plus-model AUROC there instead gives -0.315 (GC), $+0.352$ (dinucleotide) and $+0.031$ (bias-aware); at the midpoint of the two, -0.444, $+0.025$ and -0.046. Under either alternative the dinucleotide arm behaves like the bias-aware arm and not like the GC arm, so the family partition that survives every choice is the GC arm against the two others, not the composition-matched pair against the bias-aware arm. The variance ratio explains why: var(contribution) over var(baseline) is 0.61 in the dinucleotide arm against 0.09 and 0.03 in the other two, so neither axis is uncontaminated there.

These analyses do not generalise across model classes (Table 5). For the convolutional network the ordering reverses, the protocol label contributing 6.25% of variance against the baseline's 2.34%. For SpliceBERT the within-arm relation is strong in all three arms, including the bias-aware arm at

−0.568 where the 4-mer shows −0.122, so no two-family structure is apparent in that measurement. The variance attribution, the within-panel reversal and the family structure hold for the 4-mer measurement and should not be read as properties of the quantity.

Table 5: Attribution of variance in the nested contribution, and the within-arm relation between baseline and contribution, by model class ($n = 94$).

	incremental R^2		within-arm Spearman		
model	protocol | base	base | protocol	dinuc	GC	bias-aware
4-mer	1.00%	11.05%	−0.462	−0.545	−0.122
CNN	6.25%	2.34%	−0.120	−0.410	+0.075
SpliceBERT	1.07%	24.06%	−0.583	−0.767	−0.568

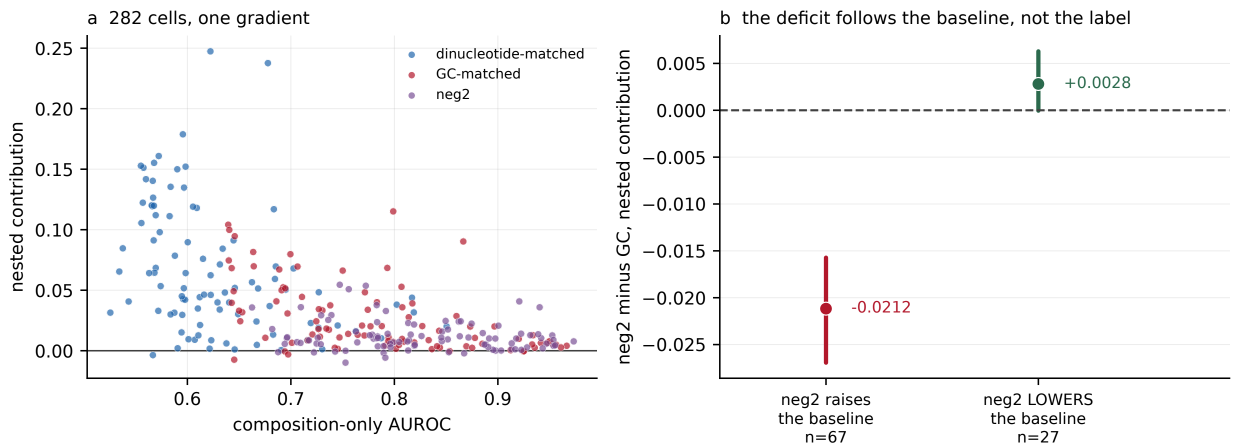

Figure 3: (a) All 282 protocol-dataset cells: nested contribution against the composition-only AUROC left by the protocol, by protocol (pooled Spearman −0.600, $p = 6 \times 10^{-29}$). (b) Datasets partitioned by whether the bias-aware protocol raised or lowered the composition baseline relative to GC matching ($n = 67$ and 27); points are stratum means and bars are 95% percentile intervals from a bootstrap over datasets (4000 draws). The contribution deficit is abolished where the baseline falls.

3.5 The span holds for three model classes

Both neural models were trained on all three arms under the same folds, seed and code path, with per-window out-of-fold scores retained (Figure 4, Table 6).

The span was present for every model class and was smallest, at 3.76-fold, for the largest model. The bias-aware protocol yielded the lowest contribution for all three model classes while carrying the highest composition baseline, so the inversion between apparent difficulty and measured contribution is not specific to a bag-of-k-mers measurement. Within a protocol the three model classes themselves span 2.65-fold (dinucleotide), 3.36-fold (GC) and 4.14-fold (bias-aware), which is the like-for-like comparison: the protocol effect is of the same order as the effect of changing model class, not

Table 6: Nested contribution under three negative-set protocols for three model classes, $n = 94$ datasets, identical rows and folds. Spans are ratios of panel means with protein-clustered percentile intervals.

model	dinucleotide	GC	bias-aware	span
4-mer	+0.0663	+0.0265	+0.0122	5.42 (4.43–6.58)
CNN	+0.0860	+0.0330	+0.0113	7.63 (6.23–9.63)
SpliceBERT	+0.1754	+0.0890	+0.0466	3.76 (3.27–4.34)
composition alone	0.6274	0.7827	0.8248	

larger than it. The two-arm contrast was +0.0398 (CI +0.0325 to +0.0477) for the 4-mer, +0.0530 (CI +0.0440 to +0.0627) for the convolutional network and +0.0864 (CI +0.0778 to +0.0951) for SpliceBERT. The bias-aware arm also reorders two of the three model classes: the 4-mer's +0.0122 there exceeds the convolutional network's +0.0113, where the network is ahead of it in both other arms. The difference is +0.0010 with a protein-clustered interval of −0.0016 to +0.0034, so what the protocol does is not so much reverse the ranking as destroy it, leaving two model classes that the other two protocols separate confidently indistinguishable. Ranking methods is what a benchmark is for, so this is the most direct cost we can show. On the ratio scale the ordering reverses, with multipliers of 3.08, 3.51 and 2.38 (geometric mean over the 77 datasets positive in both arms for all three models), so the contrast does not increase along this ladder. The three classes differ in architecture, receptive field, pretraining and parameter count at once, so this is evidence that the effect is not specific to a bag-of-k-mers measurement, and not a controlled experiment on capacity.

Decomposing the log multiplier over the 262 dataset-by-model cells in which that model was positive in both arms, a looser rule than the 77-dataset intersection above, RBP identity accounted for 64.8% of variance, cell line for 0.2% and model class for 2.8%, with 32.8% residual. These shares are not directly comparable, and with type-II sums of squares they do not sum to one: a 79-level factor absorbs 29.7% of the variance of relabelled data whereas a three-level factor absorbs 0.8%. Against a stricter null that permutes protein labels between datasets while preserving dataset-by-model blocks, and which therefore absorbs 57.6% by itself, the excess attributable to protein identity was +7.2 points ($p = 0.003$).

That excess and the direct test below are one line of evidence and not two, because the protein factor is nearly the dataset factor at 79 levels against 94, so both rest on the fifteen proteins measured in both cell lines. The direct test is that the same protein's log multiplier agreed between cell lines at $r = 0.586$ (Spearman 0.594) over 40 protein-by-model pairs. Those rows are not independent, since they come from fifteen proteins by three models sharing windows, labels and folds, so we report the protein-collapsed version as primary: averaging over models gives $r = 0.607$ ($p = 0.017$, Spearman 0.546, $p = 0.035$, $n = 15$), an order of magnitude weaker in p than the uncollapsed 0.0001 and the figure consistent with this paper's own standard of resampling proteins. Per model the agreement is $r = 0.753$ for the 4-mer and 0.772 for SpliceBERT but only 0.316 ($p = 0.32$) for the convolutional network, so it is not a property of all three. Cell line was not detectable, although with fifteen proteins measured in both lines the design has little power. Model class was small but detectable ($p = 0.023$), consistent with SpliceBERT's lower multiplier, and robust to the censoring induced by requiring both arms positive.

We do not decompose the span into a compression term and a residual protocol effect. That decomposition requires transplanting a model's discriminability increment across baselines, and its central assumption, that the increment is invariant to the baseline against which it is measured, is violated in these data. Across a grid of twelve specifications, 4 of 6 sign tests held for the 4-mer, 6 of 6 for the convolutional network and 1 of 6 for SpliceBERT; the log-odds link reverses the sign entirely, which is the behaviour Mood [18] describes for odds ratios under unobserved heterogeneity. With three arms the residual's sign follows the direction of transplant, +0.0452 carrying an increment from a high-baseline arm to a low-baseline one and −0.0259 in reverse. We report the raw contrast, which requires no transplant, link or transportability assumption.

This subsection carries two provenance limitations. For 20 of the 94 datasets in the dinucleotide arm the retained neural scores were produced under a partition that is not chromosome-grouped. Fold sizes were preserved, so count-based checks do not detect it. Up to 23 chromosomes appear in a single fold, and up to 44.5% of rows have a same-strand neighbour within 1 kb assigned to a different fold. All 564 score sets across the three arms are now audited: the GC and bias-aware arms are clean, 94 of 94 each, with at most five chromosomes per fold and no cross-fold neighbour at all. The 4-mer is refit on the study folds for every dataset and is unaffected. Restricted to the 74 correctly partitioned datasets the contrasts were +0.0436, +0.0506 and +0.0878, and the spans 6.02, 8.33 and 3.92, every one of them inside the protein-clustered interval reported for the full panel. We report the full panel as primary throughout and these as a sensitivity, because the defect is conservative: dropping the affected datasets widens the span instead of narrowing it. On the affected datasets themselves the defect is not small. The difference in differences, comparing each neural model's margin over the 4-mer between arms, is +0.0294 for the convolutional network and +0.0110 for SpliceBERT between leaky and clean datasets, which for the convolutional network is as large as its whole GC-arm contribution; the leaky datasets are also the panel's largest, with a median of 15,596 pairs against 4,373, so that figure confounds the defect with size and is an upper bound.

The second limitation is one of scale. The 4-mer entered the nested fit as a log odds and the neural models as probabilities; logistic regression is not invariant to a nonlinear transform of a covariate. Refitting the neural arms on the logit scale increased their contributions by +0.0008 (CNN) and +0.0030 (SpliceBERT) on the GC arm and moved the contrasts by at most 0.0004, so the reported values are conservative. AUROC itself is invariant to monotone transforms.

3.6 One structural feature of the calibration replicates on an externally constructed benchmark

We applied the same decomposition to the negative sets released by Horlacher et al. [4], using their positives, peak calling, negative-set construction and fold assignments, over the 45 datasets shared

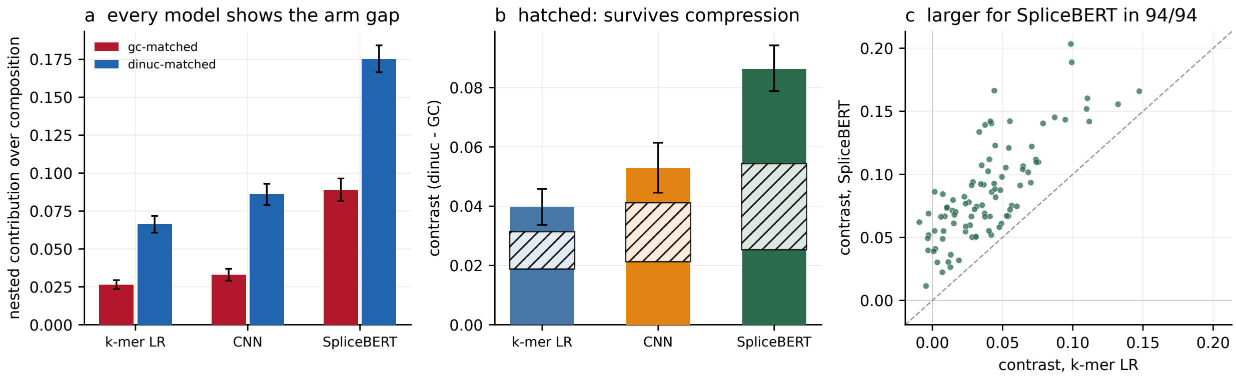

Figure 4: Three model classes on identical rows and folds, $n = 94$ datasets: a 4-mer logistic regression, a 7089-parameter convolutional network and a fully fine-tuned 19.78 M-parameter SpliceBERT. (a) Nested contribution in each composition-matched arm, per model; bars are standard errors over datasets. (b) The two-arm contrast per model, with 95% percentile intervals from a bootstrap over the 79 proteins taking all datasets of each sampled protein (4000 draws). The hatched band spans the protocol effect obtained by transplanting a discriminability increment in either direction, +0.0188 to +0.0314 for the 4-mer, +0.0213 to +0.0412 for the convolutional network and +0.0253 to +0.0543 for SpliceBERT: every member keeps the sign, so the contrast is not an artefact of AUROC compression, but its width is why we report the raw contrast and decline the decomposition in the text. (c) The contrast per dataset, SpliceBERT against the 4-mer; it is larger for SpliceBERT in all 94.

with our panel (Table 7, Figure 5). The benchmark is external in its construction and not in its biology: the underlying experiments overlap ours, so this tests whether the same measurement behaves the same way when somebody else builds the negatives, not whether it behaves the same way on new proteins. Their released sets also carry no model-only AUROC, so difficulty there can only be read from the composition baseline, which is a weaker proxy than the one we use internally.

Table 7: The same decomposition applied to the released negative sets of Horlacher et al. [4], $n = 45$ datasets. The 4-mer's own AUROC is not available for their negative sets, so difficulty on their benchmark is read from the composition baseline.

negative set	composition	composition + 4-mer	contribution
negative-1 (transcript background)	0.8211	0.8606	+0.0395
negative-2 (other RBPs' sites)	0.7560	0.7726	+0.0166

The span on their benchmark was 2.381, against 2.17 for the corresponding family-matched pair in our own data, the GC-versus-bias-aware comparison. Their negative-2 is constructed like our bias-aware arm, which makes that the appropriate comparison and not our dinucleotide-versus-GC span of 2.50.

The two-family structure of the preceding subsection reproduced arm for arm. The within-arm relation between baseline and contribution was -0.645 ($p = 1.8 \times 10^{-6}$) for their composition-matched transcript-background arm and -0.187 ($p = 0.22$) for their other-RBPs'-sites arm, against

−0.545 and −0.462 for our two composition-matched arms and −0.122 for our bias-aware arm. Two things did not replicate, and both belong here and not in the Limitations. The pooled relation across their two arms, the external counterpart of our −0.600, is −0.202 ($p = 0.056$, $n = 90$), which is null at their sample size. And the direction of the level relation is reversed: their negative-2 has both the lower baseline and the lower contribution, so on their benchmark apparent difficulty and measured contribution move in the same direction rather than in opposite directions. A cross-family residual is consistent with that and a single-family headroom relation is not, but the ordering was not predicted in advance and we do not present it as a confirmation.

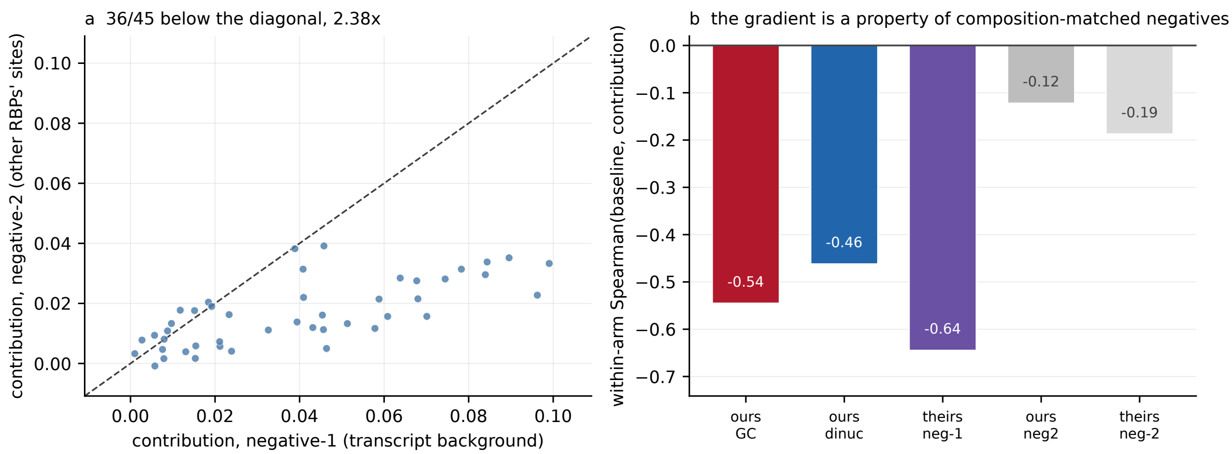

Figure 5: Replication using the released negative sets, peak calling and fold assignments of Horlacher et al. [4], over the 45 datasets shared with our panel; only the measurement is ours. (a) Nested contribution under their negative-2 against their negative-1, paired by dataset; 36 of 45 lie below the diagonal. (b) Within-arm Spearman correlation between the composition baseline and the contribution, for the composition-matched arms of both benchmarks and the other-RBPs'-sites arms of both. The relation is present in one group and absent in the other, independently in each benchmark.

3.7 Reporting the baseline helps in sample but does not replicate out of sample

We tested the recommendation in the situation it is intended for, a reader holding two studies that used different protocols and wishing to compare what the models contributed (Figure 6).

On our data, normalising by headroom improved cross-protocol rank agreement in 3 of 3 protocol pairs and reduced disagreement on a common scale in 3 of 3, by 58%, 10% and 48%. Rank agreement is scale-free and cannot be flattered by normalisation. This held for all three model classes, with mean changes in rank agreement of +0.040 (4-mer), +0.094 (convolutional network) and +0.061 (SpliceBERT).

The improvement is weaker than it looks. Only the GC-versus-dinucleotide improvement has an interval excluding zero (+0.051, CI +0.007 to +0.115); the remaining two pairs are consistent in direction and individually null, and the pair that reaches significance is the one in which the protocol label contributes 0.05% of variance. That interval is marginal, not comfortable, and it does not survive adjustment for the three protocol pairs: the Bonferroni-corrected 98.33% interval is −0.003 to +0.137. The supporting sign evidence is weaker than it looks too, since three of three pairs has $p = 0.125$ under an exchangeable null, and the nine cells over three pairs by three model classes are not nine independent successes because they are drawn from the same 94 datasets and three arms. The test also fails to replicate out of sample, which matters more. On the 45 datasets of Horlacher et al. [4], rank agreement fell from 0.706 to 0.656 under normalisation and disagreement rose from 0.860 to 0.908, which are the two criteria we specified in advance as falsifying. The change in rank agreement was −0.050 (CI −0.222 to +0.140), so at $n = 45$ this is a failure to replicate rather than a refutation.

We therefore recommend reporting the composition-only AUROC obtained under the same protocol alongside every headline AUROC, on the grounds that it is the only summary of the protocol's effect that a reader can act on, and we do not recommend treating any normalisation as rendering contributions comparable across protocols.

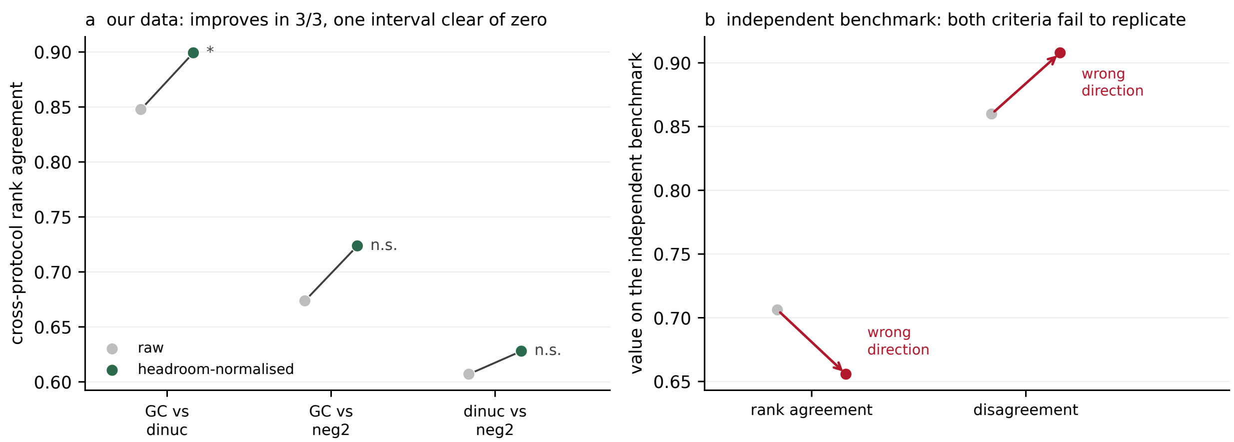

Figure 6: (a) Cross-protocol rank agreement between nested contributions before and after division by the headroom of the baseline, for each of the three protocol pairs in our data. Agreement improves in 3 of 3 pairs; only the GC-versus-dinucleotide improvement has a protein-clustered interval excluding zero. (b) The same test on the independent benchmark of Figure 5, $n = 45$: rank agreement falls and disagreement rises under normalisation, the two criteria specified in advance as falsifying. The change in rank agreement is −0.050 (−0.222 to +0.140), a failure to replicate rather than a refutation.

4 Discussion

How much a sequence model adds over sequence composition is not a property of the model alone. Holding the model, the positives, the folds and the estimator fixed, and varying only how the negative windows were assembled, the measured contribution of a 4-mer varied 5.4-fold, and between 3.8 and 7.6-fold across three model classes. The direction of that variation is opposite to the direction of apparent performance: the protocol that made the discrimination hardest by AUROC produced the largest contribution, and the protocol that made it easiest produced the smallest.

This has consequences for reporting practice. The composition-only AUROC obtained under the same protocol should be given alongside every headline AUROC, which costs one additional logistic regression on nineteen features. And contributions measured under different negative-set protocols should not be compared, which withdraws a comparison instead of offering one, and is the more consequential of the two, because cross-study comparison of this kind is what a benchmark is normally used for.

Apparent difficulty is therefore not a proxy for the stringency of an evaluation. The bias-aware protocol is well motivated as a means of removing compositional and accessibility artefacts, and its design is not at fault; what does not follow is the inference from a lower or higher score to how much a model has learned. Separating one protein's sites from another's is largely a composition task, so a protocol that poses that task leaves a sequence model comparatively little to add. Two things make it a composition task, and only one of them is about binding sites. Distinct RBPs occupy distinct transcript regions, and the base composition characteristic of each region is readable from nineteen features; distinct RBPs also occupy compositionally distinct sites within a region. Rebuilding the bias-aware arm with the donor draw stratified on region removes the first and leaves the second: it costs the arm 0.020 AUROC of baseline, which is 46% of its excess over the GC arm, and a quarter of its measured contribution, and leaves its position among the three protocols unchanged. The lesson for a designer is that a protocol which frees a covariate the other protocols hold fixed is not simply a harder version of them, and that the comparison to make before attributing a baseline to biology is against the same protocol with that covariate restored.

The relation between the composition baseline a protocol leaves and the contribution a model subsequently shows was strong for composition-matched negatives, both in our data and independently in that of Horlacher et al. [4], and absent for negatives drawn from the sites of other proteins. A residual difference between the two groups survived adjustment for the baseline. On the 4-mer measurement this points to what the negative window is, more than to how tightly it is matched: contributions appear comparable within a group of protocols that draw negatives from the same kind of object, and less so between such groups. Three qualifications apply, and together they are why we present this as a feature of one measurement. It was recovered from the 4-mer and did not reproduce for either neural model. The statistic is not invariant to which term is placed on its horizontal axis, and under the two alternatives the dinucleotide arm joins the bias-aware arm and not the GC arm, so the partition that survives every choice separates the GC arm from the other two. And on Horlacher et al. [4]'s benchmark the pooled relation is null.

Protocol dependence can be reduced by fitting a transformation but not by choosing one from among the standard set. Eight standard reparameterisations reduced the span without closing it, and the smallest of them, division by the baseline's headroom, is the pair of AUROCs we recommend reporting, expressed as a single number. An exponent that closes the span does exist, but it does not transport. The exponent fitted to our three protocols leaves an independent benchmark essentially where it started; the exponent fitted to that benchmark is 2.4 times ours; and the exponent differs by a factor of 1.22 between model classes measured on the same data. This is the sense in which no protocol-free measure of contribution is available. The obstacle is not that no transformation works but that a transformation which works has to be refitted for each measurement, which is precisely what a comparable quantity cannot require.

The composition baseline used here stops at order two, matching what a dinucleotide-preserving shuffle would hold fixed. That is defensible, and it is a choice, so we measured what it costs. Refitting the baseline at order three, adding the 63 trinucleotide frequencies, removed approximately four fifths of the 4-mer's measured contribution and 87% of the two-arm contrast on 30 size-stratified datasets, and on the bias-aware arm the median dataset retained nothing. For a bag-of-k-mers model, most of what is reported as contribution over a composition baseline is one further order of composition, not motif recognition or any positional feature. Any study reporting an increment over a composition baseline is reporting a quantity indexed by the order of that baseline, and few state which order was used.

The collapse is bounded in two ways. Protocol dependence is not an artefact of where the baseline stops: at an order-three baseline the two-arm contrast remained +0.0067, +0.0309 and +0.0545 for the three model classes at $n = 94$, all with intervals excluding zero. The collapse is also specific to models whose features are nested within the span of the baseline. Over a trinucleotide baseline the 4-mer's contribution was positive in 65 of 94 datasets against 94 of 94 for SpliceBERT, and after accounting for the reduced headroom the raised baseline leaves, the 4-mer lost 1.41 times what SpliceBERT lost in the dinucleotide arm and 1.29 times in the GC arm. A benchmark meant to separate model classes, instead of ranking them by how much short-range composition they absorb, could therefore raise the order of its baseline. That recommendation carries a circularity worth stating: trinucleotide counts are linear aggregates of 4-mer counts, so an order-three baseline is structurally unfavourable to k-mer models, and separates models by how much of their signal lies outside short-range composition, not by capacity as such.

Two features of the order-three numbers are cautions and not results. The span across protocols does not collapse at order three but grows, to 7.16, and it grows because the bias-aware denominator falls to +0.0015 with only 17 of 30 datasets positive, which is the same near-zero-denominator pathology that disqualifies normalising by the excess over chance; its interval runs to 22.6 and it should not be read as a magnitude. And the surviving order-three contribution is concentrated, with 51% of the positive mass in the GC arm sitting in 3 of the 30 datasets.

Horlacher et al. [4] established that negative-set construction changes apparent performance across many RBP prediction methods and proposed the bias-aware alternative. The present work does not contest that and depends on it, since their released negative sets provide our only external validation. What we add is a calibration of it. What the protocol moves is not just apparent AUROC but the increment over an explicit baseline, and it moves that increment in the opposite direction, by a factor of between 3.8 and 7.6 depending on the model class, and no reparameterisation renders that increment comparable across benchmarks. On this panel the protocol effect is of the same order as the effect of changing model class, which is what these benchmarks exist to resolve: within a protocol the three model classes span 2.65 to 4.14-fold, against 3.8 to 7.6-fold across protocols within a model. The same dependence has since been reported for transcription factor binding sites [5], on a different assay, a different class of model and without an explicit baseline, which suggests the problem is not specific to eCLIP. The circularity we cite in the Introduction for variant-effect prediction has the same shape. Here it is measured for one estimand in one assay, with the arithmetic part separated from the part that is not.

Limitations

Every magnitude reported here is indexed to a mono- and dinucleotide baseline. The protocol dependence is not so indexed.

The relation in the title holds throughout our own data, across three arms and three model classes, and does not hold on the one independent benchmark we tested. On Horlacher et al. [4]'s two negative sets the harder arm by both AUROCs in Table 7 also yields the smaller contribution, so difficulty and contribution move together there. Their two arms belong to different protocol families, which is the structure that would predict it, but that was not a prediction we made in advance.

Achieved matching is not exact matching. The GC matcher's operative acceptance bound is 0.15 and not its nominal 0.05, holding the nominal tolerance for 94.8% of pairs, and the dinucleotide matcher reduces composition distance 2.27-fold, to a median L_1 of 0.220, so degrees-of-freedom statements are exact for the design and approximate for the data.

All three model classes were measured under all three protocols, but the study covers two cell lines and one assay. Whether the calibration holds for other crosslinking protocols or other organisms is untested. The transform sweep, the baseline attribution and the recommendation test are reported for all three model classes. The order-three analysis is reported for all three model classes at $n = 94$ on the two composition-matched arms, and for the 4-mer additionally on 30 size-stratified datasets across all three arms; the two use different bootstrap schemes, protein-clustered for the first and over datasets for the second. No model larger than 20 M parameters was tested, and no published RBP-specific method: the three classes are a capacity ladder, not a survey of the methods a benchmark would rank.

Two limitations concern the provenance of the neural scores. For 20 of 94 datasets in the dinucleotide arm the retained scores came from a partition that is not chromosome-grouped, and we report the full panel as primary with the 74 correctly partitioned datasets as a sensitivity; every restricted value lies inside the interval reported for the full panel, and the restriction widens the spans instead of narrowing them. Initialisation was unseeded across all 2830 fold-runs, at the cost quantified in Methods.

Positive sets differ slightly between arms. Window construction is identical, but a positive is dropped when its matcher finds no acceptable negative and the matchers fail on different windows, so the two composition-matched arms' positive sets have a Jaccard similarity with median 0.9972 and minimum 0.9237 and are identical in 10 of 94 datasets. The negatives are therefore almost, but not exactly, the only difference between arms.

We applied no significance filter of our own to the positives, and none was available: ENCODE's released eCLIP peak files are already thresholded, and across our 95 files the minimum log2 fold-enrichment is 3.0000 and the minimum $-\log_{10} p$ is 3.0003, matching the criteria of Van Nostrand et al. [3], with 91.7% of the 463,091 peaks carrying ENCODE's IDR label and nine K562 datasets contributing non-IDR peaks. There were therefore no weak peaks to exclude. What we do discard is peak *width*: every positive is a fixed 101 nt window on the peak midpoint, so a 20 nt peak and a 2 kb peak are treated identically, and for the wider peaks the window may not contain the crosslink site. No expression filter was applied to negatives; that question is bounded by the control reported above and not removed. Region class is assigned from the window midpoint under a fixed precedence, with introns computed as gaps between exons and no gene-type filter, so a window spanning an annotation boundary can be labelled either way. This affects a matching covariate and not an outcome, and it is the same annotation for both classes in the two arms that match on it.

The verification harness asserts values and not the provenance of the code that produced them, and the fold-partition defect above is of exactly the class that this distinction permits: it was identified by inspection and by none of the 696 assertions, as was the region asymmetry of Section 3.2. Its manuscript audit checks only values written to three or more decimal places, and requires a decimal point, so bare integers are not checked at all; those were checked by hand, and three were wrong in an earlier draft of this manuscript.

Confidence intervals on a single dataset's contribution are Wald intervals on the DeLong standard error and not bootstrap intervals, and the Firth [17] coefficient intervals reported in the accompanying repository resample rows, which is narrower than gene-level clustering would give.

5 Conclusion

How much a sequence model adds over sequence composition is a joint property of the model and of the negative set it was evaluated against, and on this panel the two axes move against each other: the protocol that made the discrimination easiest by AUROC was also the one that left a sequence model least to add.

For a benchmark designer this has two operational consequences. Report the composition-only AUROC obtained under the same protocol alongside every headline AUROC; it costs one logistic regression on nineteen features and it is the only summary of a protocol's effect that a reader can act on. And do not compare contributions measured under different negative-set protocols. That restriction is one we can justify, where a normalisation that would remove the need for it is not: eight standard reparameterisations reduce the span without closing it, and the exponent that does close it is fitted, not chosen, differing between benchmarks and between model classes measured on the same data.

A third consequence bears on how protocols are compared with each other. The three here hold different things fixed, and the one that frees transcript region pays for it in a baseline that a nineteen-feature model can read. Loosening a constraint does not make a protocol more stringent.

Data availability

All primary data are public. Positive windows derive from ENCODE eCLIP narrowPeak files [1, 3]; every dataset used, with its ENCFF file accession and ENCSR experiment accession, is listed in Supplementary Table S1 (95 datasets, 79 proteins, 95 experiments). Gene annotation is GENCODE v45 [9]. Cell-line expression used in the transcription control comes from four ENCODE polyA RNA-seq quantification files: ENCFF286KKZ and ENCFF829LCN for K562 (ENCSR115PIZ), and ENCFF863QWG and ENCFF376IXQ for HepG2 (ENCSR245ATJ and ENCSR813BDU). The external validation uses the negative sets and fold assignments released by Horlacher et al. [4] at doi:10.5281/zenodo.10600977, downloaded and md5-verified.

Derived data supporting every number in this paper, including per-window out-of-fold model scores for all three model classes and all per-dataset result tables, are released with the code under CC BY 4.0. Intermediate window tables containing genomic sequence are not redistributed; they are regenerated from the accessions above by the pipeline in one command.

Supplementary Table S1 lists all 95 panel datasets. One of them, NCBP2 in K562, did not clear the out-of-fold pair floor in the GC arm, so it carries no three-protocol results; its `in_three_arm_panel` field is false and its result columns are empty.

Code availability

All analysis code is available at `https://github.com/NirmalKumar31/rbp-protocol-calibration` under the MIT licence, with `results/` and `data/evidence/` under CC BY 4.0. Every published value is reproducible offline from the repository with `python scripts/verify.py --local results/tables`, which runs 696 numeric assertions against a frozen expectations file and asserts that no fewer than that number ran, so a check cannot silently skip. One quantity is an exception and is marked as such in the text: the design effect, whose block bootstrap needs the genomic coordinates that the published per-window score files do not carry. It is reproduced by `scripts/design_effect.py` against the window store, which is regenerated from the accessions above in one command.

Use of AI tools

Generative AI (Anthropic Claude) was used as a coding assistant to implement the analysis pipeline and to assist in drafting text, which the author revised. The study was conceived, designed and directed by the author, who made every analytical and interpretive decision, determined which claims the evidence supports, and is solely responsible for the content of this work. All reported values are produced by code in the accompanying repository and are reproduced by its verification harness; no reference was generated by a language model.

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Competing interests

The author declares no competing interests.

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